

Natural Language Processing

Laboratory

Ex1
 $D_1 = \text{"cat eats fish"}$ $[1 \ 0 \ 1 \ 1 \ 0]$

$D_2 = \text{"dog eats fish"}$ $[0 \ 1 \ 1 \ 1 \ 0]$

$D_3 = \text{"cat likes fish"}$ $[1 \ 0 \ 0 \ 1 \ 1]$

Ex2

$$tf(\text{cat}, D_1) = \frac{1}{3}$$

$$tf(\text{eats}, D_1) = \frac{1}{3}$$

$$tf(\text{fish}, D_1) = \frac{1}{3}$$

$$\sum tf(t, D_1) = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$$

Ex3

$$n = 3$$

$$df(\text{cat}) = 2$$

$$df(\text{dog}) = 1$$

$$df(\text{eats}) = 2$$

$$df(\text{fish}) = 3$$

$$df(\text{likes}) = 1$$

$$idf_{\text{cat}} = \log \frac{3}{2}$$

$$idf_{\text{dog}} = \log \frac{3}{1}$$

$$idf_{\text{eats}} = \log \frac{3}{2}$$

$$idf_{\text{fish}} = \log \frac{3}{3}$$

$$idf_{\text{likes}} = \log \frac{3}{1}$$

Ex 4

$$\text{idf cat} = \log \frac{3}{2} \quad \Rightarrow \quad \text{tf-idf cat} = \frac{1}{3} \cdot \log \frac{3}{2}$$

$$\text{tf cat} = \frac{1}{3}$$

~~$$\text{tf-idf dog} = \frac{1}{3}$$~~

~~$$\text{tf-idf cats} = \frac{1}{3} \cdot \log \frac{3}{2}$$~~

~~$$\text{tf-idf fish} = \frac{1}{3} \cdot \log \frac{3}{1}$$~~

Ex 5

$$\cos(x, y) = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|}$$

$$= \frac{1 \cdot 1 + 1 \cdot 1}{\sqrt{3} \cdot \sqrt{2}} = \frac{2}{\sqrt{3} \cdot \sqrt{2}} = \frac{2}{\sqrt{6}}$$

Ex 6

1) D_1 a similarity correlation

2) D_2 similarity by 1 thay nhai

3) Term a idf thay nhai "medical"



Part C

Prediction 1 $\rightarrow$ vocabulary blocking 300,000 từ

Prediction 2 $\rightarrow$ Sparsity

Ans 1: Thừa (sparse)

Ans 2: từ 0 zero entries blocking 90%

Prediction 3 $\rightarrow$ Search

không đúng ví dụ

"Đáp án câu này phải là một"

"Đáp án câu này không là một"

